

Week 2

Complex numbers - A1

Example ① $2x - 8 = 0$ $x = 4$
one solution

② $P(x) = x^2 - 25 = 0$ $x = 5, -5$
two solutions

③ $P(x) = x^2 + 1 = 0$ (No real solutions)
No solutions

Let i be a "imaginary" number
such that $i^2 = -1$

Def a complex number: $z = a + bi$
 $a, b \in \mathbb{R}$

$\mathbb{C}$ = all complex numbers

Ex

$$z_1 = 3 - 2i$$

$$z_2 = 4 + 5i$$

Def $z_1 = a_1 + b_1 i$, $z_2 = a_2 + b_2 i$

$$z_1 = z_2 \Leftrightarrow a_1 = a_2, b_1 = b_2$$

$$z = a + bi$$

$\text{Re}(z) = a$ real part of z

$\text{Im}(z) = b$ imaginary part of z

Ex

$$z = 7 - 8i \quad \text{Re}(z) = 7$$
$$\text{Im}(z) = -8$$

Operations

① addition

$$z_1 = a_1 + b_1 i$$

$$z_2 = a_2 + b_2 i$$

$$z_1 + z_2 = (a_1 + a_2) + (b_1 + b_2) i$$

Ex $(2-7i) + (3+8i) = 5 + i$

Note a real number is a complex number with imaginary part zero
 $\mathbb{R}$ is a subset $\mathbb{C}$

Complex Conjugate

$$z = a + bi \quad \bar{z} = a - bi$$

Ex $z = -4 - 7i \Rightarrow \bar{z} = -4 + 7i$

if $z = a$ (if $\text{Im}(z) = 0$)

$$\bar{z} = a = z$$

$$\overline{(\bar{z})} = z$$

multiplication

$$(a+b)(c+d) = ac + ad + bc + bd$$

$$(a_1 + bi)(a_2 + b_2i) = a_1a_2 + a_1b_2i + a_2b_1i + b_1b_2i$$

$$(but\ i^2 = -1)$$

$$= (a_1a_2 - b_1b_2) + (a_1b_2 + a_2b_1)i //$$

$$i^4 = (i^2)^2 = (-1)^2 = 1$$

$$i^3 = i^2 \cdot i = -1$$

Ex $(3-2i)(4+i) = 12 + 3i - 8i + 2 = 14 - 5i //$

Special case:

$$(3+4i)(3-4i) = 9 - \cancel{12i} + \cancel{12i} + 16 \\ = 25$$

$$(a+bi)(a-bi) = a^2 - abi + abi + b^2 \\ = a^2 + b^2$$

Def $Z = a + bi$

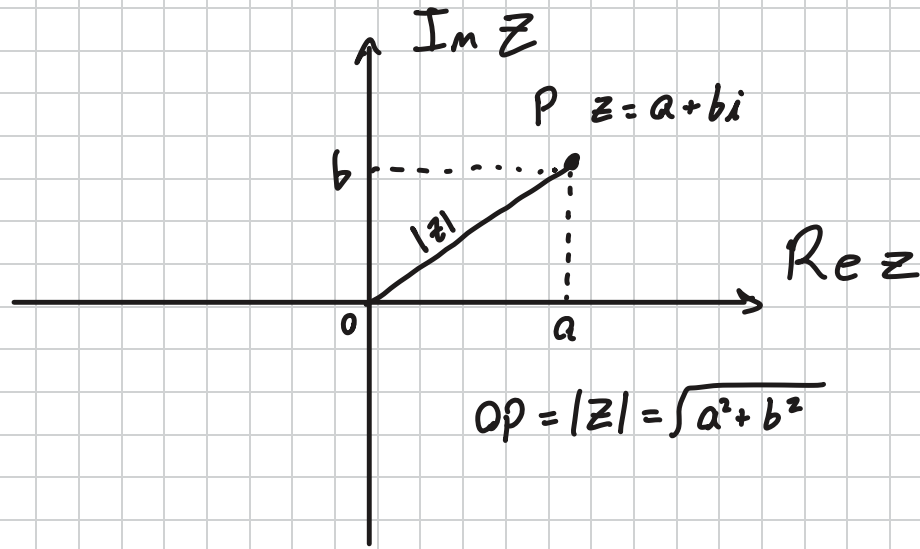
absolute
value of Z

$$|Z| = \sqrt{a^2 + b^2}$$

always a positive real number

$$Z \cdot \bar{Z} = |Z|^2, \quad |Z| = |\bar{Z}|$$

the complex plane



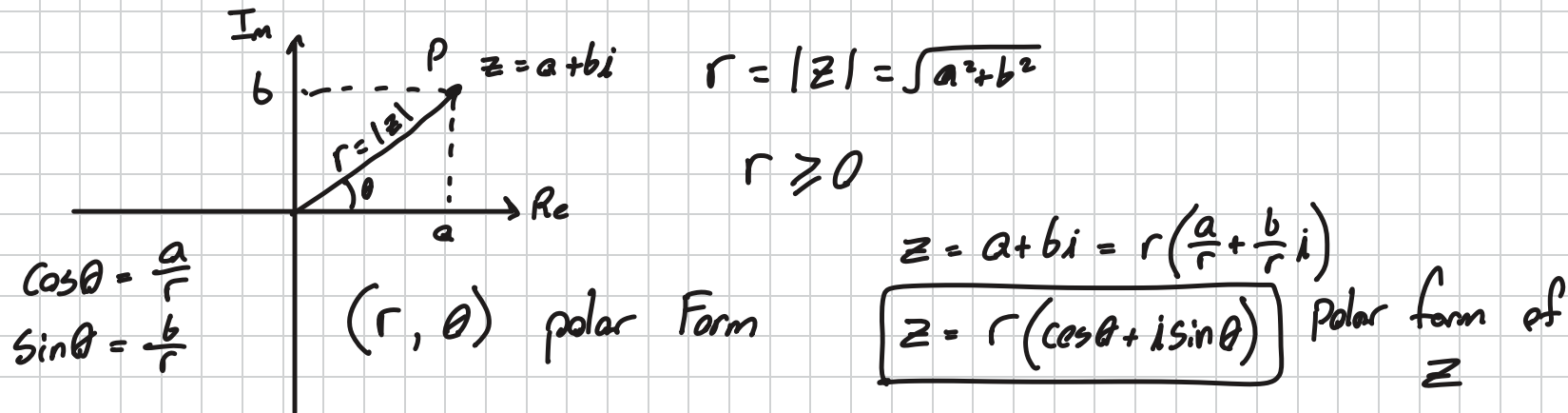
Division

$$\begin{aligned} \frac{3+4i}{2-i} &= \frac{3+4i}{2-i} \times \frac{2+i}{2+i} = \frac{(3+4i)(2+i)}{(2-i)(2+i)} = \frac{6+3i+8i-4}{5} = \frac{2+11i}{5} = \frac{2}{5} + \frac{11}{5}i \end{aligned}$$

→ multiply top and bottom by complex conjugate of the denominator.

$z = a + bi$ Cartesian form of a complex number

The polar Form



E_x

$$z = 1 + i$$

$$r = |1 + i| = \sqrt{2}, \quad \theta = \frac{\pi}{4}$$

$$z = \sqrt{2} \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} i \right) = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$$

polar form of a complex number

$$Z = a + bi \quad \text{Cartesian form}$$

$$r = |z| = \sqrt{a^2 + b^2}$$

$$Z = r (\cos \theta + i \sin \theta) \quad \text{polar form}$$

$$\cos \theta = \frac{a}{r}, \quad \sin \theta = \frac{b}{r}$$

θ - argument of z .

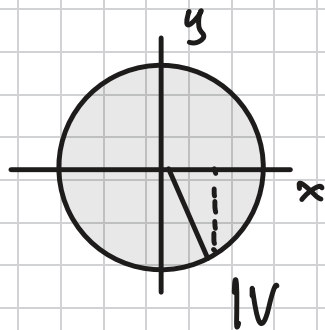
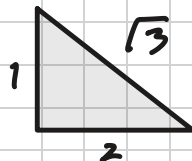
Example

$$z = 1 - \sqrt{3}i$$

$$\Rightarrow r = 2 = |z|$$

$$z = 2\left(\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)$$

$$\Rightarrow \begin{aligned} \cos\theta &= \frac{1}{2} \\ \sin\theta &= -\frac{\sqrt{3}}{2} \end{aligned}$$



$$\theta = -\frac{\pi}{3} \quad \theta = \arg(z)$$

$$\Rightarrow z = 2\left(\cos\left(-\frac{\pi}{3}\right) + i\sin\left(-\frac{\pi}{3}\right)\right)$$

Note
Arg of z is
not unique

We define the principal argument

$$\text{Arg}(z) = \arg(z) = 0$$

Such that $-\pi < \theta < \pi$

Ex

$$\text{Arg}(z) = -\frac{\pi}{3}$$
$$\arg(z) = -\frac{\pi}{3} \text{ or } \frac{5\pi}{3} \text{ or } \dots$$

Consider

$$z_1 = r_1 (\cos \theta_1 + i \sin \theta_1)$$
$$z_2 = r_2 (\cos \theta_2 + i \sin \theta_2)$$

$$① - \sqrt{3}i$$

$$|z| = \sqrt{a^2 + b^2}$$

$$= \sqrt{1^2 + (\sqrt{3})^2}$$

$$= \sqrt{1 + 3}$$

$$= 2$$

$$z_1 \cdot z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$$

Exercise: prove formula above

$$|z_1 z_2| = |z_1| |z_2|$$

$$\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$$

$$(\cos \theta + i \sin \theta)^n = (\cos n\theta + i \sin n\theta)$$

de Moivre's theorem.

Example calculate $(1+i)^{10}$

① write $z = 1+i$ in polar form

$$z = \sqrt{2} \left(\underbrace{\frac{1}{\sqrt{2}}}_{\cos \theta} + i \underbrace{\frac{1}{\sqrt{2}}}_{\sin \theta} \right)$$

$$z = 1+i = \sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$$

$$(1+i)^{10} = (\sqrt{2})^{10} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)^{10}$$

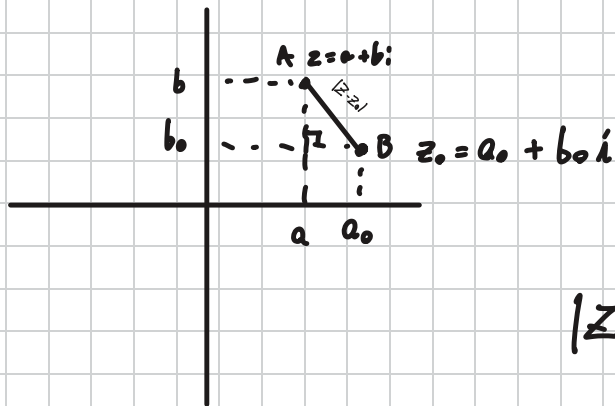
$$= 2^5 \left(\cos \frac{5\pi}{2} + i \sin \frac{5\pi}{2} \right)$$

$$= 2^5 i$$

$$= 32i$$

Describe all complex numbers

$$z \text{ such that } |z - 1 + i| = \sqrt{2}$$



$$|AB| = \sqrt{(a-a_0)^2 + (b-b_0)^2}$$

$$z - z_0 = (a-a_0) + (b-b_0)i$$

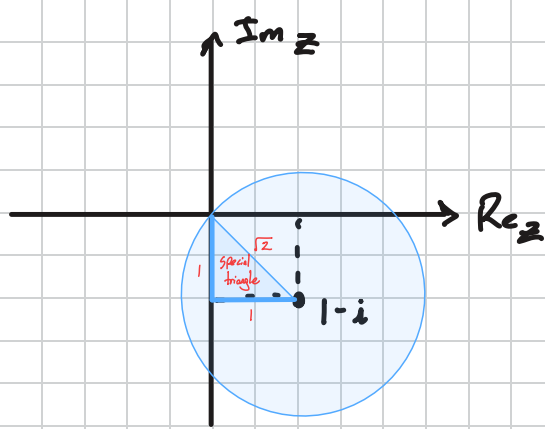
$$|z - z_0| = \sqrt{(a-a_0)^2 + (b-b_0)^2} = |AB|$$

$$|z - 1 + i| = |z - (1 - i)|$$

$$\text{Let } 1 - i = z_0$$

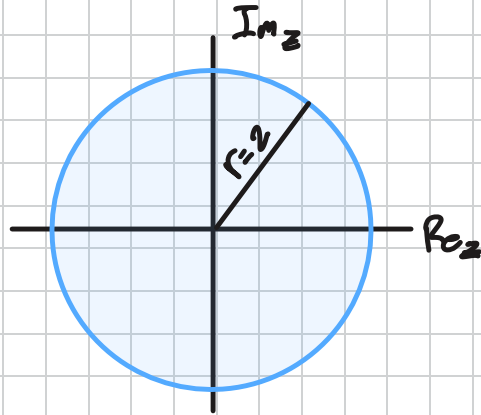
$$|z - z_0| = \sqrt{2}$$

$\Rightarrow z$ must be on a circle
of radius $\sqrt{2}$ centered at $1 - i$



Ex describe the point z in
Complex plane such that $|z|=2$

Circle of $r=2$
center at origine



Roots of complex numbers

Example $z^2 + 1 = 0 \Rightarrow z^2 = -1$

two roots $z_1 = -i$, $z_2 = i$

We try solving equations of the type

$$z^n = w \quad \begin{array}{l} w \in \mathbb{C} \text{ given} \\ n \text{ integer} \end{array}$$

Step 1 write w in polar form

$$w = r(\cos \theta + i \sin \theta)$$

Solve $z^n = r(\cos \theta + i \sin \theta)$

it will have n roots:

$$Z_0 = r^{\frac{1}{n}} \left(\cos \frac{\theta}{n} + i \sin \frac{\theta}{n} \right)$$

Exercise: verify that $Z_0^n = w$

$$Z_1 = r^{\frac{1}{n}} \left(\cos \frac{\theta + 2\pi}{n} + i \sin \frac{\theta + 2\pi}{n} \right)$$

$$Z_{n-1} = r^{\frac{1}{n}} \left(\cos \frac{\theta + 2\pi(n-1)}{n} + i \sin \frac{\theta + 2\pi(n-1)}{n} \right)$$

What if we keep going?

$$Z_n = r^{\frac{1}{n}} \left(\cos \frac{\theta + 2\pi n}{n} + i \sin \frac{\theta + 2\pi n}{n} \right)$$

$$\downarrow = r^{\frac{1}{n}} \left(\cos \left(\frac{\theta}{n} + 2\pi \right) + i \sin \left(\frac{\theta}{n} + 2\pi \right) \right)$$

$$= r^{\frac{1}{n}} \left(\cos \frac{\theta}{n} + i \sin \frac{\theta}{n} \right) = z_0$$

All n roots are equally spaced and lie on a circle of radius $r^{\frac{1}{n}}$ centered at the origin

Example find all the cubic roots of $-8i$

$$z^3 = -8i = 8(0-i)$$

$$\cos \theta = 0$$

$$\sin \theta = -1$$

$$\theta = \frac{3\pi}{2}$$



$$Z^3 = 8 \left(\cos \frac{3\pi}{2} + i \sin \frac{3\pi}{2} \right)$$

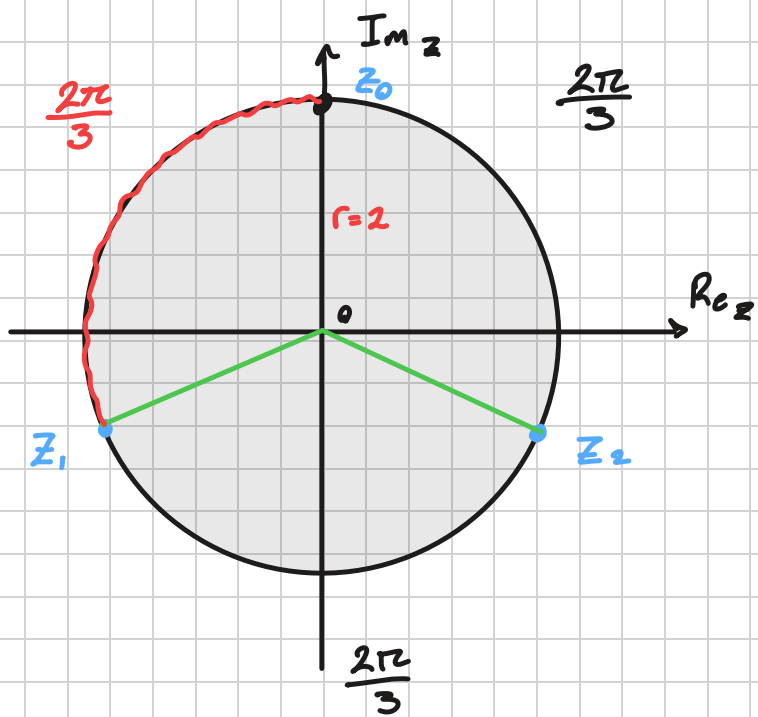
$$Z_0 = 8^{\frac{1}{3}} \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right) = 2i$$

$$Z_1 = 8^{\frac{1}{3}} \left(\cos \frac{\frac{3\pi}{2} + 2\pi}{3} + i \sin \frac{\frac{3\pi}{2} + 2\pi}{3} \right)$$

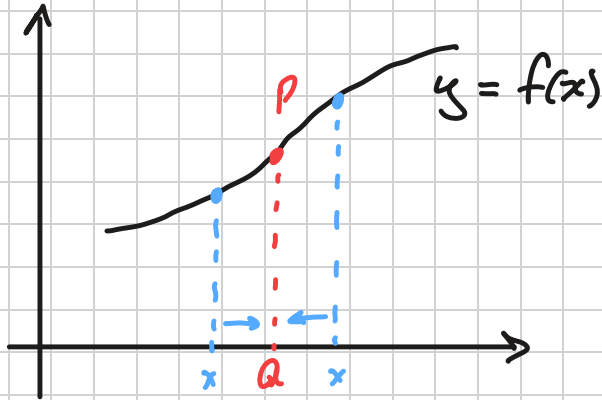
$$= 8^{\frac{1}{3}} \left(\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6} \right) = 2 \left(-\frac{\sqrt{3}}{2} - \frac{1}{2}i \right) = -\sqrt{3} - i$$

$$Z_2 = 8^{\frac{1}{3}} \left(\cos \frac{\frac{3\pi}{2} + 4\pi}{3} + i \sin \frac{\frac{3\pi}{2} + 4\pi}{3} \right)$$

$$= 2 \left(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6} \right) = 2 \left(\frac{\sqrt{3}}{2} - \frac{1}{2}i \right) = \sqrt{3} - i$$



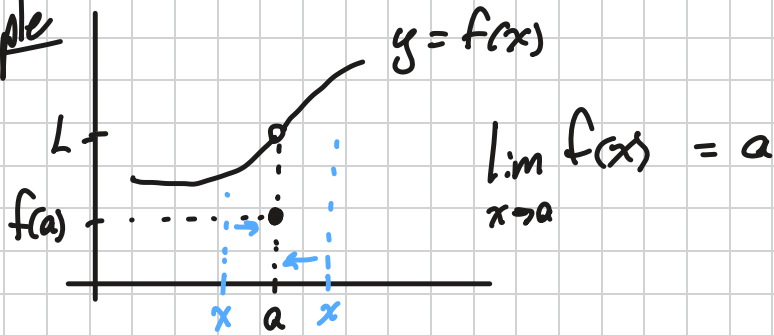
Limits of functions



if f is "nicely behaved" (continuous)

$$\lim_{x \rightarrow a} f(x) = f(a)$$

Example



Example

$$f(x) = \frac{x^2 - 9}{x - 3}$$

$$D(f) = \mathbb{R} - \{3\}$$

$$\text{if } x \neq 3 \quad f(x) = \frac{\cancel{(x-3)}(x+3)}{\cancel{(x-3)}} = x+3$$

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = \lim_{x \rightarrow 3} (x + 3) = 6$$

